Predictive Maintenance – HVAC Units

Standard Operating Procedure (SOP)

Purpose

This SOP defines the mandatory procedure for responding to predictive maintenance alerts associated with abnormal temperature and power consumption in HVAC units (HVAC_1, HVAC_2).

The objective is to ensure safe, consistent maintenance actions that restore normal operating conditions and prevent equipment failure.

Scope and Applicability

This SOP applies to HVAC units (HVAC_1, HVAC_2) installed in Buildings B1 and B2.

It must be followed by authorised maintenance personnel whenever predictive alerts are generated by the AI Engine and distributed via the M365 Copilot interface.

Roles and Responsibilities

- **Maintenance Technician** – Responsible for executing inspection and corrective steps in accordance with this SOP.
- **Electrical Authorised Person** – Responsible for all electrical inspections and interventions.
- **Site Supervisor** – Responsible for validation of results and signoff of work completion.

Overview

This SOP describes the standard process used to inspect, diagnose, and correct airflow or electrical performance issues in HVAC units (HVAC_1, HVAC_2) located across Buildings B1 and B2.

Trigger Conditions (Entry Criteria)

This SOP is initiated when all of the following conditions are met:

- Sustained internal HVAC temperature exceeds 25°C.
- Electrical power draw exceeds 22 kW for a sustained duration.
- A predictive alert is issued by the AI Engine.

Health & Safety Rules

Before beginning work, personnel must adhere to the following safety protocols:

- Wear standard site personal protective equipment (PPE), which includes safety glasses, steel-toe boots, and a high-visibility vest.
- Utilize Class 0 electrical safety gloves rated for 1000V when interacting with electrical components.
- Ensure a Lockout/Tagout (LOTO) kit is on hand and actively utilized to isolate power prior to inspection.

Only personnel certified for electrical work may inspect or interact with electrical components. Nonauthorised personnel must not access electrical contactors or live panels under any circumstances.

Required Tools

- A digital multimeter that has been calibrated within the last 12 months.
- A thermal imaging camera.
- Standard insulated screwdrivers.
- Fresh MERV-13 air filters for replacement.

Step-by-Step Execution Process

1. **Preparation:** Start by verifying the details of the work ticket and the AI summary on your mobile device through the M365 Copilot interface.
2. **Locating the Asset:** Find the exact HVAC unit utilizing the building ID, floor number, and room details specified in the initial alert.

3. **Power Isolation:** Perform standard LOTO procedures on the unit's primary local disconnect switch to safely cut off power.
4. **Internal Inspection:** Take off the primary access panel to visually examine both the blower motor and the main filter bank.
5. **Filter Replacement:** If you observe heavy particulate buildup, replace the air filters, as this is a primary cause of high temperatures and restricted airflow.
6. **Electrical Check:** Inspect the unit's main electrical contactor. Look closely for any pitting or burn marks that might explain the anomalous power readings.
7. **System Restart:** Remove the LOTO devices, turn the power back on, and initiate a manual test cycle.
8. **Observation:** Monitor the HVAC unit for 5 minutes. Check the telemetry dashboard to verify that both the power and temperature metrics have stabilized and returned to normal baseline ranges.

Exit Criteria

This SOP may be closed only when:

- HVAC internal temperature has returned to within normal operating range.
- Electrical power draw has returned to baseline levels.
- The unit has completed a minimum 5-minute monitored test cycle.
- All readings have been recorded and verified.

Exception Handling and Escalation

If temperature or power consumption does not normalise:

- Do not release the unit into full operation.
- Escalate to the Site Supervisor or Reliability Engineering team.
- Document persistent faults and initiate further diagnostics if required.

Post-Job Audit and Verification

All findings, replaced components, and measurements must be logged in the D365 mobile application.

A digital or physical signature from the responsible supervisor is required. This sign-off confirms that exit criteria have been met and triggers audit review and AI model feedback.

SOP Governance

- **SOP Owner:** Facilities & Reliability Engineering
- **Version:** 1.0
- **Approval Authority:** Facilities Manager
- **Effective Date:** [Insert Date]
- **Review Cycle:** Annual or upon system updates